One-Dimensional Random Walk

Szymon Birban
University of Warsaw, Faculty of Physics

October 2026

1. Objective

The aim of this experiment is to simulate independent one-dimensional random walks and verify two theoretical predictions: the Gaussian distribution of particle positions and the linear growth of the mean squared displacement (MSD).

2. Model and Theory

Consider N independent particles starting at $x_i(0) = 0$. Their positions evolve according to

$$x_i(t+1) = x_i(t) + \xi_i(t), \qquad \xi_i(t) \sim \mathcal{N}(0,1). \tag{1}$$

All increments are independent. After t steps,

$$x_i(t) = \sum_{k=0}^{t-1} \xi_i(k). \tag{2}$$

Since a sum of independent Gaussian random variables is Gaussian, and

$$\mathbb{E}[x_i(t)] = \sum_{k=0}^{t-1} \mathbb{E}[\xi_i(k)] = 0, \tag{3}$$

$$\mathrm{Var}[x_i(t)] = \sum_{k=0}^{t-1} \mathrm{Var}[\xi_i(k)] = t, \tag{4}$$

the particle position follows

$$\boxed{x_i(t) \sim \mathcal{N}(0,t).} \tag{5}$$

Its probability density is therefore

$$\rho(x,t) = \frac{1}{\sqrt{2\pi t}} \exp\left(-\frac{x^2}{2t}\right), \qquad t > 0. \tag{6}$$

The mean squared displacement is estimated numerically as

$$\widehat{\mathrm{MSD}}(t) = \frac{1}{N} \sum_{i=1}^{N} [x_i(t) - x_i(0)]^2. \tag{7}$$

Because $\mathbb{E}[x_i(t)] = 0$, the theoretical prediction is

$$\boxed{\mathrm{MSD}(t) = \mathbb{E}[x_i^2(t)] = t.} \tag{8}$$

This agrees with the diffusion relation $\mathrm{MSD}(t) = 2Dt$, where the diffusion coefficient is $D = 1/2$ in the chosen units.

3. Results and Discussion

The model was implemented in Python using NumPy and Matplotlib. Independent Gaussian increments were generated for each particle and time step. Particle positions were obtained through cumulative summation using `np.cumsum`. Position distributions were estimated using normalized histograms, while MSD was computed by averaging squared displacements over all particles at each time step.

3.1. Random Walk Trajectories

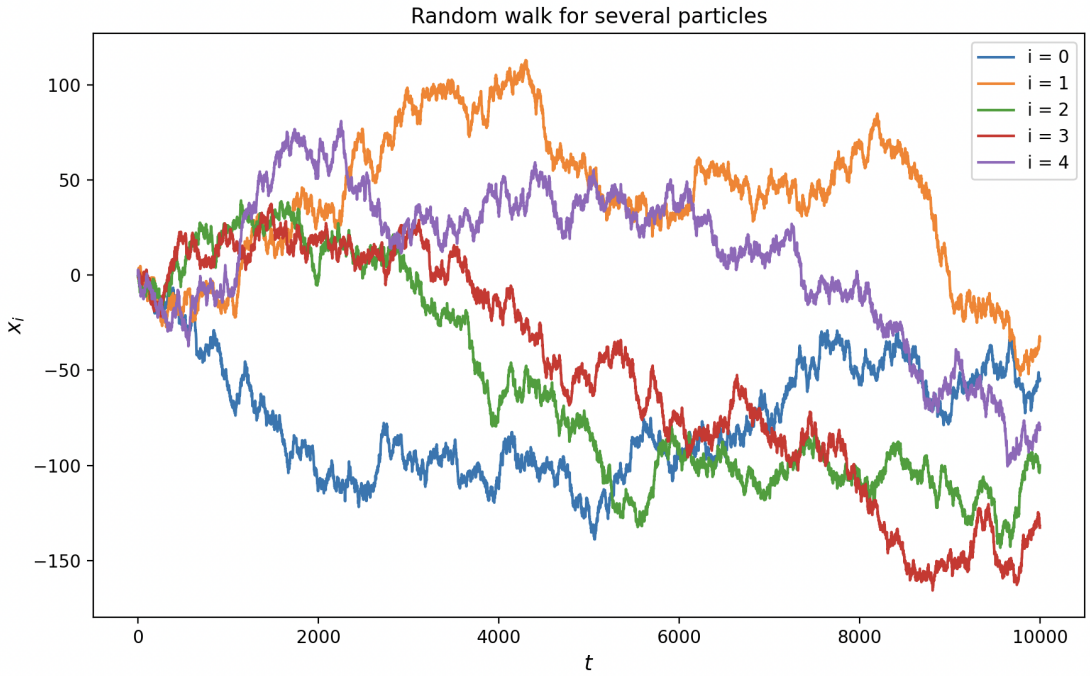


Figure 1: Selected trajectories of independent one-dimensional random walkers.

Figure 1 shows five independent random walk trajectories over 10,000 time steps. Although all particles start at the origin and their displacements follow the same probability distribution, their trajectories evolve differently. No systematic drift is imposed by the model, as the expected displacement at each step is zero. Nevertheless, individual particles can reach large positive or negative displacements due to accumulated random fluctuations.

3.2. Position Distribution

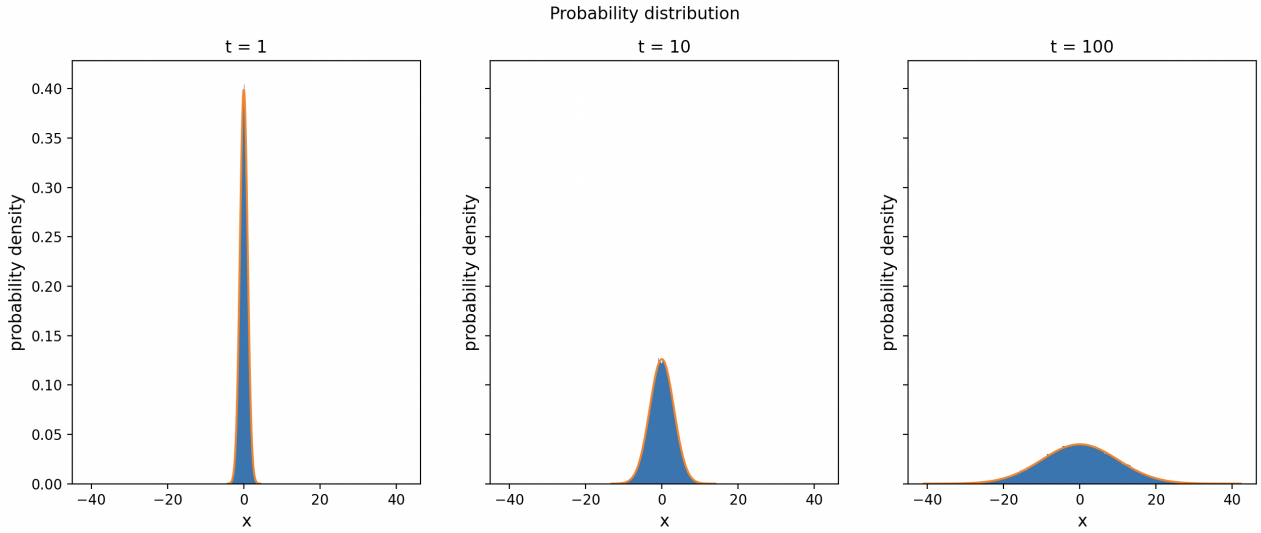


Figure 2: Probability distributions of particle positions at different t and analytical Gaussian distributions.

Figure 2 shows the probability distributions of particle positions at different t, compared with the analytical Gaussian distributions. The numerical results closely agree with theoretical predictions. As time increases, the distributions become wider and their peaks decrease, consistent with the theoretical variance $\sigma_x^2(t) = t$. This illustrates the diffusive spreading of particles over time.

3.3. Mean Squared Displacement

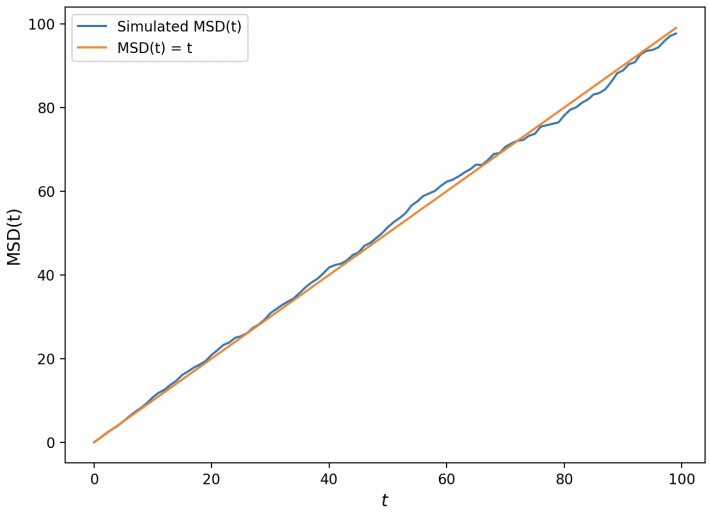


Figure 3 compares the simulated mean squared displacement (MSD) with the theoretical prediction (MSD(t) = t). The numerical results show approximately linear growth and closely follow the analytical solution, confirming the expected diffusive behaviour. Small deviations from the theoretical line arise from statistical fluctuations associated with the finite number of simulated particles.

4. Conclusions

The numerical simulations successfully reproduced the main properties of one-dimensional random walks. Individual particle trajectories exhibited irregular motion, while their position distributions closely followed the theoretical Gaussian predictions.

The mean squared displacement increased approximately linearly with time, in agreement with the analytical result $\mathrm{MSD}(t) = t$, confirming normal diffusion.

Small deviations between numerical and theoretical results can be attributed to statistical fluctuations arising from the finite number of simulated particles. Overall, the experiments demonstrate how predictable statistical behaviour emerges from independent random processes.